CRYPTO & WEB3 PORTFOLIO ANALYSIS · WORKED EXAMPLE

Synthetic Portfolio Review

A complete full-review example: IPS-lite, concentration, risk, liquidity, scenarios, decision record and a visual pack.

As of: 2026-10-30 14:00 UTC

Base currency: USD

Portfolio: $100,000 synthetic

Benchmark: BTC

Mode: Full Review

Data coverage: 92%

$100,000	65%	7.1%	92%
SYNTHETIC VALUE	TOP-TWO WEIGHT	DAILY CVAR	DATA COVERAGE

What this demonstrates: mandate before numbers, evidence classes, risk clusters rather than tickers, exit capacity, scenario and reverse stress, and a decision record with falsifiers. Every chart is synthetic and every material number is labelled.

SYNTHETIC — NOT A REAL PORTFOLIO

> **Important:** This is an illustrative example, not investment, tax, or legal advice. It contains no real wallet, person, or live price. The skill is read-only and never executes a transaction.

Evidence classes: VERIFIED · REPORTED · ESTIMATED · MODELED · UNVERIFIED

As of: 2026-10-30 14:00 UTC

Base currency: USD

Data coverage: 92% (four live positions; small-cap price six hours old)

Benchmark: BTC
Mode: Full Review

> Synthetic example. No real wallet, person, or investment recommendation.

1. Executive Summary

- **Main finding:** 65% of the portfolio sits in BTC and ETH, so it behaves like a high-beta crypto index with a small illiquid tail.
- **Why it matters:** The portfolio trails BTC; the gap must be separated into selection, beta, fees, and liquidity risk.
- **Verified:** Prices from two sources; balances reconciled within 0.4%.
- **Not verified:** True exit depth for the small-cap; cost basis is missing.
- **Decision:** Review the small-cap and cap the single-position weight. This is not a trade instruction.
- **What would change the view:** Small-cap exit time under five days, or beta above 1.4.

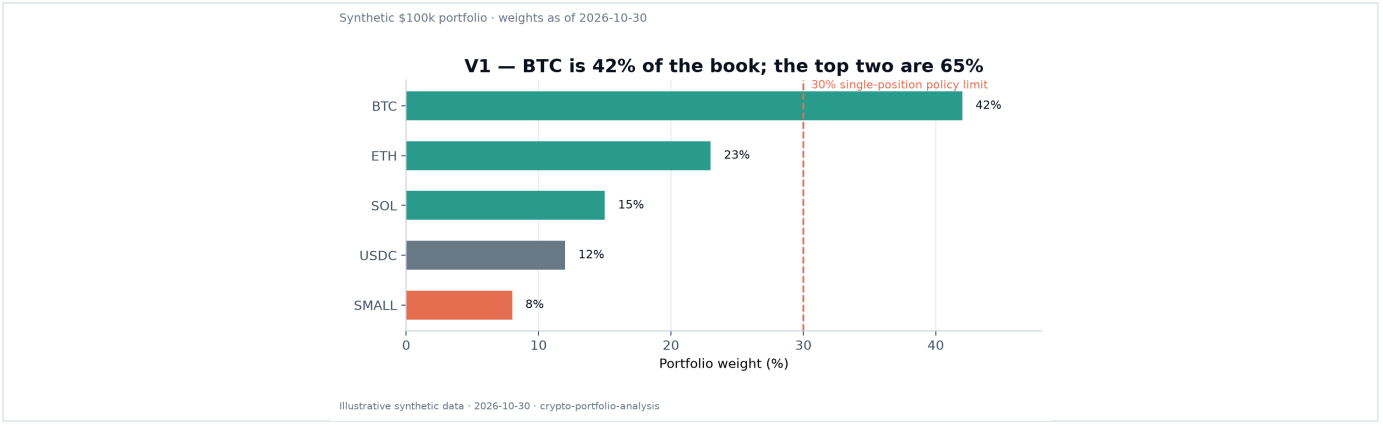
2. Mandate and IPS-lite

Item	Detail
Purpose	Growth over three years
Return objective	Long-term growth with BTC-like risk, not maximum yield
Risk tolerance	Maximum 35% drawdown
Liquidity	$20,000 within 12 months
Constraints	30% single-position cap; no leverage
Review	Quarterly; monthly risk check

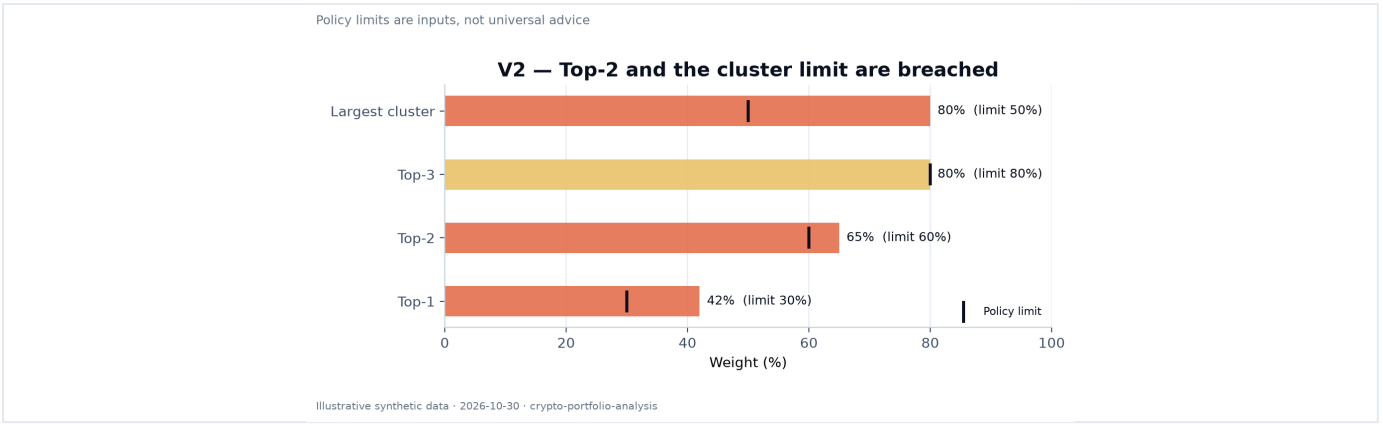
3. Portfolio Snapshot

Position	Value	Weight	Source/time	Evidence
BTC	$42,000	42.0%	Two venues, 14:00 UTC	VERIFIED
ETH	$23,000	23.0%	Two venues, 14:00 UTC	VERIFIED
SOL	$15,000	15.0%	Venue + aggregator	VERIFIED
USDC	$12,000	12.0%	Issuer + market	VERIFIED
SMALL	$8,000	8.0%	One DEX, 08:00 UTC	ESTIMATED
Total	**$100,000**	**100%**	—	—

Top-1 / Top-2 / Top-3: 42.0% / 65.0% / 80.0%
HHI: 0.2726; **effective positions:** 3.67
Cluster map: BTC 42%, ETH 23%, SOL 15%, stablecoin 12%, small-cap 8%.



Finding: BTC is 42% of the book; the top two positions are 65%. **Source/date:** Synthetic data, 2026-10-30. **Limitation:** Illustrative only; not a live portfolio.

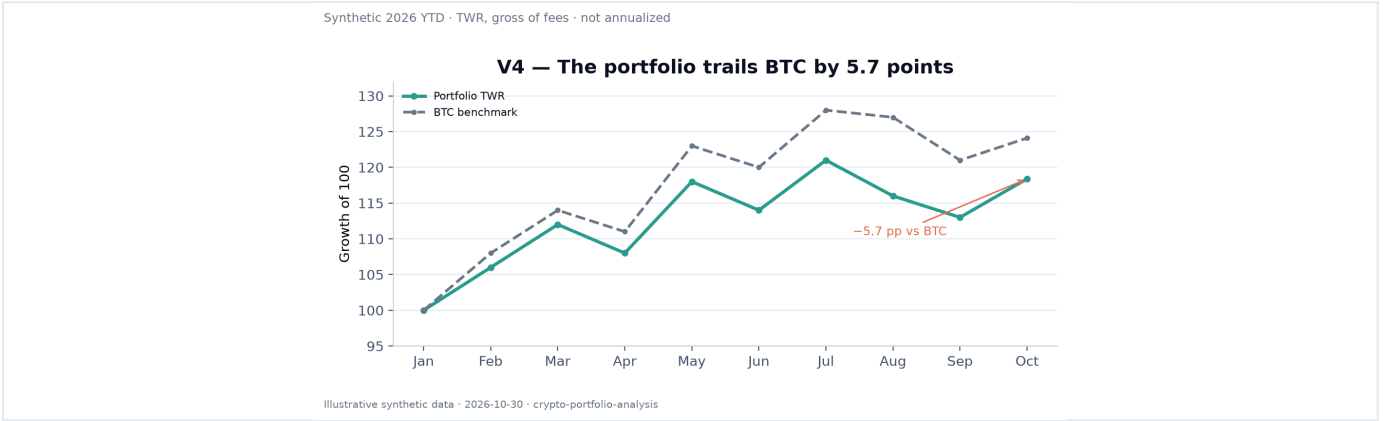


Finding: The top-two and major-crypto cluster limits are both breached. **Source/date:** Synthetic policy limits, 2026-10-30. **Limitation:** Policy limits are illustrative, not the reader's mandate.

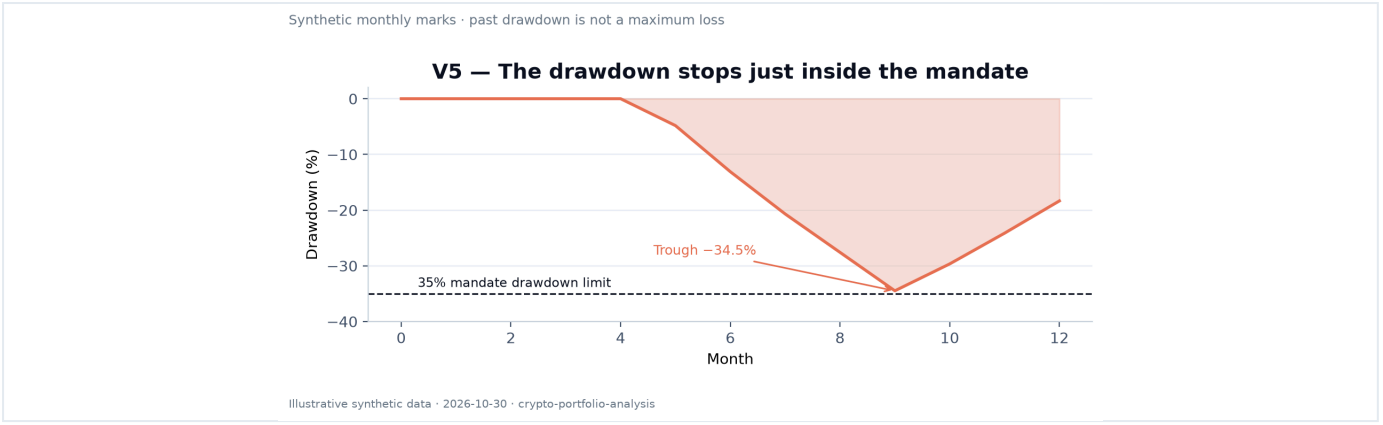
4. Performance and Attribution

Measure	Value	Window	Benchmark	Evidence
TWR	+18.4%	2026-01-01 to 2026-10-30	BTC +24.1%	VERIFIED
MWR	+16.9%	Same	—	ESTIMATED
Contribution	BTC +9.8; ETH +5.1; SOL +4.2; other −0.7	Same	—	ESTIMATED

The portfolio trails BTC by 5.7 points. With no cost basis, this is a value-change analysis, not P&L.



Finding: The synthetic portfolio trails BTC by 5.7 percentage points. **Source/date:** Synthetic price and flow data, 2026-01-01 to 2026-10-30. **Limitation:** Illustrative returns; no cost basis or fees.

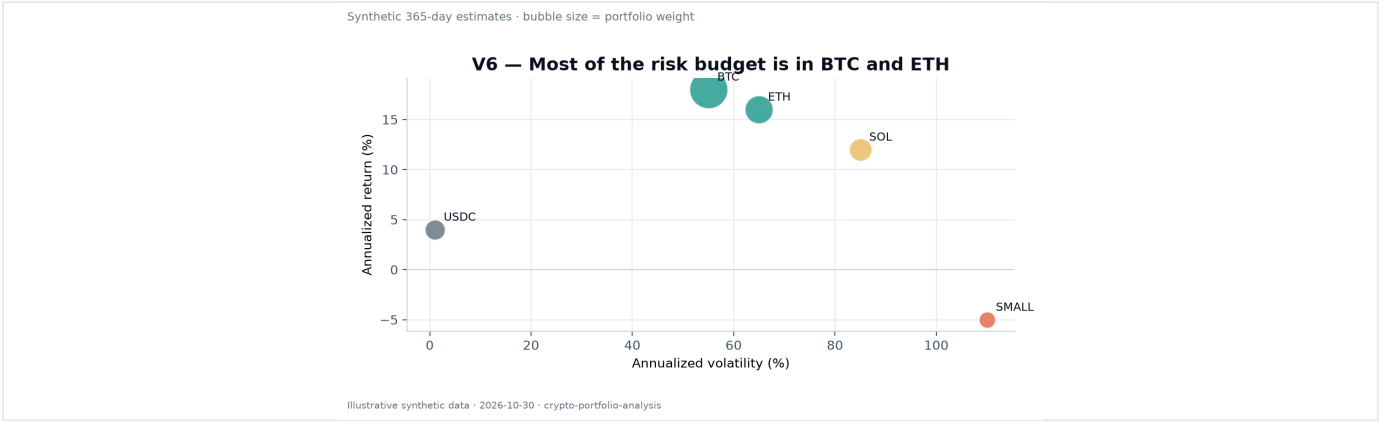


Finding: Maximum drawdown stops just inside the 35% loss limit. **Source/date:** Synthetic daily marks, 2024-10-30 to 2026-10-30. **Limitation:** A backtest is not a forecast of the next drawdown.

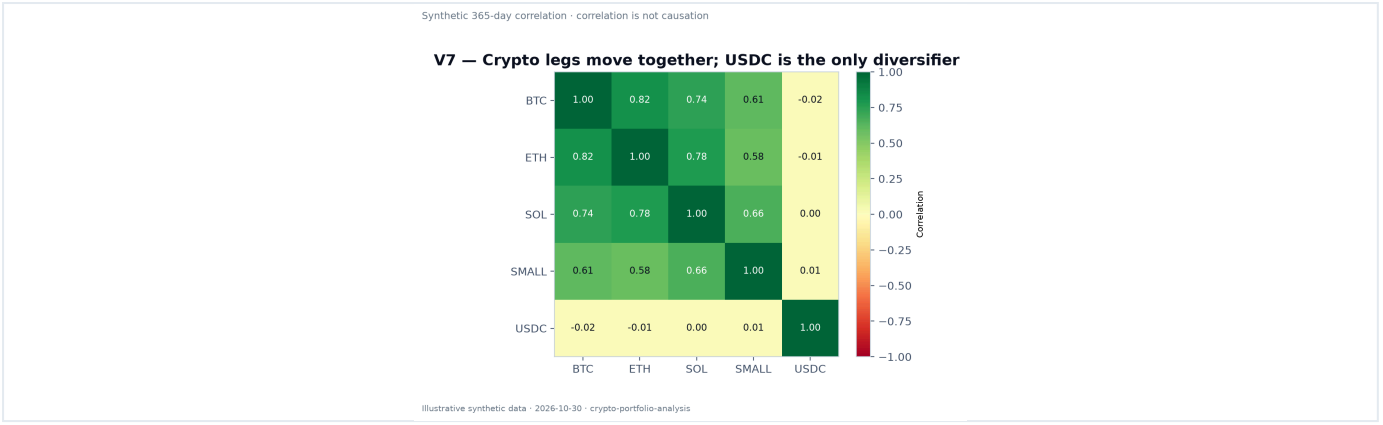
5. Risk Dashboard

Measure	Value	Window	Method	Limit
Volatility	58%	365d	Historical	50%
Max drawdown	−34%	2 years	Daily marks	−35%
VaR 95% 1d	4.8%	365d	Historical	4.0%
CVaR 95% 1d	7.1%	365d	Historical	6.0%
Beta vs BTC	1.12	365d	OLS	1.20
Top risk contribution	BTC 46%	365d	Component CVaR	—

VaR is not a maximum loss. The portfolio is close to its drawdown limit and above its volatility and CVaR limits.



Finding: BTC and ETH carry most of the risk budget; the small-cap adds a fat tail. **Source/date:** Synthetic returns, 365 days. **Limitation:** Volatility and VaR are model-dependent.

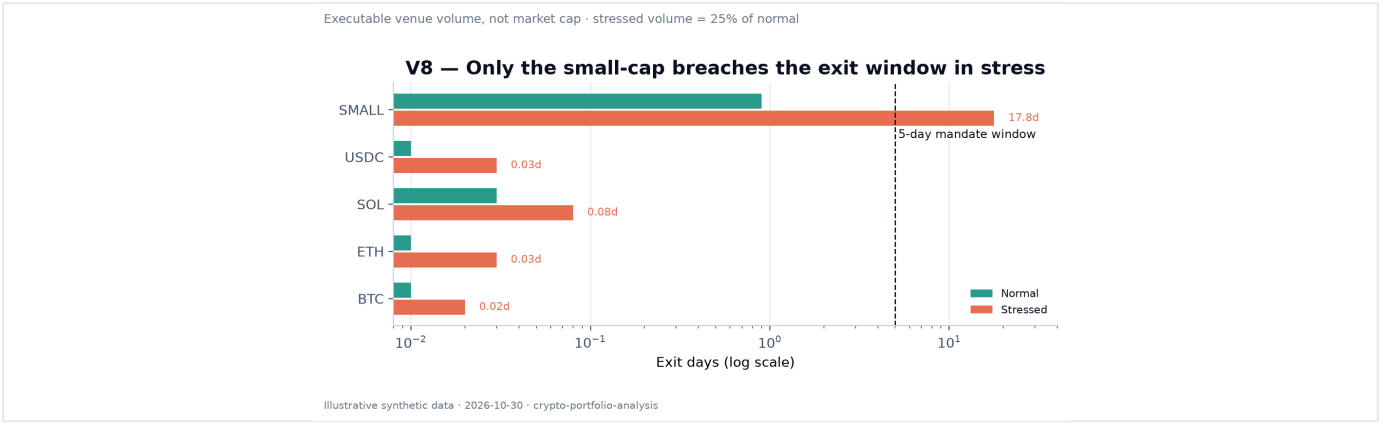


Finding: Crypto legs are highly correlated; USDC is the main price diversifier. **Source/date:** Synthetic daily returns, 365 days. **Limitation:** Correlation is unstable and can rise in a crisis.

6. Liquidity and Exit

Position	Value	Daily executable volume	5% exit days	Stressed exit days	Gate
BTC	$42,000	$8.0B	0.01	0.01	None
ETH	$23,000	$4.0B	0.01	0.01	None
SOL	$15,000	$600M	0.01	0.03	None
USDC	$12,000	$2.0B	0.01	0.01	Issuer redemption
SMALL	$8,000	$180,000	0.9	17.8	None

The small-cap is not a liquidity problem at 8% of this portfolio, but it is the first position that cannot be exited quickly in stress.



Finding: Only the small-cap breaches the five-day exit window under stress. **Source/date:** Synthetic volume data, 2026-10-30. **Limitation:** Volume is a proxy, not a full order-book simulation.

7. Asset-Level Findings

BTC

Reserve asset; deep liquidity; no cash flow. Main risks: regulation, custody, beta.

ETH

Smart-contract infrastructure; staking yield and burn; validator/MEV and L2 value migration risks.

SOL

High-performance L1; staking and network usage; outage history, validator concentration, competition.

USDC

Liquidity buffer, not cash. Issuer, reserve, and redemption-gate risk. A 12% weight is a policy choice, not a risk-free allocation.

SMALL

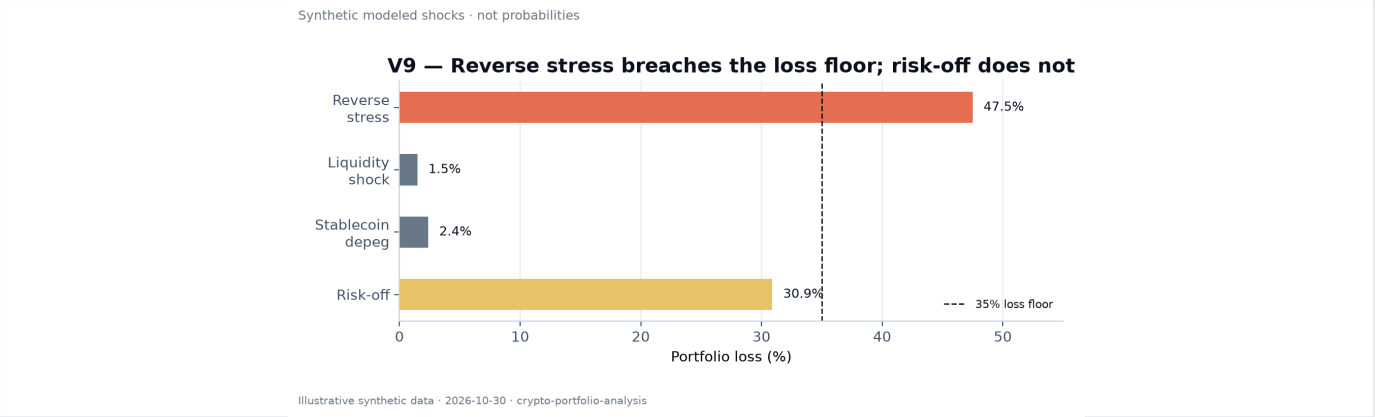
Low float, one DEX price, no independent second source. Tokenomics and unlock schedule need verification. Treat the position as `ESTIMATED`, not verified.

8. DeFi and On-chain

No lending or LP positions in this example. Recent BTC/ETH exchange outflows are mildly negative; this is context, not a thesis. Entity labels are probabilistic.

9. Scenarios

Scenario	Shocks	Portfolio loss	Worst position	First breach
Risk-off	BTC −30%, ETH −40%, SOL −50%, SMALL −70%	$30,900 / 30.9%	SMALL	Near drawdown limit
Stablecoin depeg	USDC −20%	$2,400 / 2.4%	USDC	Liquidity gate
Liquidity shock	Small-cap volume −75%	17.8 exit days	SMALL	Not the 12-month need
Reverse stress	BTC −45%, ETH −55%, SOL −65%, SMALL −85%	$47,500 / 47.5%	SMALL	35% drawdown breach



Finding: The reverse-stress path breaches the 35% loss floor; risk-off does not. **Source/date:** Synthetic shocks, 2026-10-30. **Limitation:** Shocks are illustrative, not probabilities.

10. Alternatives and Rebalancing

Alternative	Risk	Liquidity	Concentration	Cost/tax	What would make it wrong?
Current	High beta	Good except small-cap	Top-2 65%	Low	BTC rally continues
Risk-reduced	Lower beta	Better	Top-2 55%	Medium	Bull market missed
Diversified-within-thesis	Similar beta	Medium	Top-2 50%	Medium	Correlations rise

Bands: ±5 percentage points; ±3 for the small-cap.
Tax note: Cost basis missing; tax impact not assessable.

11. Decision Record

- **Decision:** Review the small-cap; enforce the 30% single-position cap; classify USDC as a counterparty-bearing liquidity buffer.
- **Rationale:** Top-2 65%, weak small-cap liquidity, underperformance vs BTC.
- **Size/implementation:** No size without constraints.
- **Falsifiers:** Small-cap exit time below five days, or beta above 1.4.
- **Review:** 2027-01-30; monthly liquidity and beta check.

12. Sources and Limitations

- Prices: two independent venues/aggregators, 2026-10-30 14:00 UTC.
- Balances: wallet plus second source; 0.4% difference.
- Small-cap: one DEX, 08:00 UTC; no independent confirmation.
- Cost basis missing; value change is not P&L.
- Not investment, tax, or legal advice.

Appendix A — Reproducibility and quality checks

This appendix is part of the worked example. It shows the checks a reviewer should expect before accepting a full review, not a claim that the synthetic inputs are live.

Check	Result in this example
Mandate captured before the verdict	IPS-lite table on page 2: growth, three-year horizon, 35% drawdown limit, $20k liquidity need, 30% single-position cap.
Data coverage stated	92%; the small-cap price is six hours old and has one venue. It is marked ESTIMATED.
Evidence classes	Every table uses VERIFIED, REPORTED, ESTIMATED, MODELED or UNVERIFIED; no material number is silently filled.
Concentration and clusters	Top-1 42%, top-2 65%, top-3 80%; major-crypto cluster 80%, so the ticker count overstates diversification.
Risk budget	Volatility 58%, max drawdown −34%, VaR(95) 4.8%, CVaR(95) 7.1%, beta 1.12; volatility and CVaR exceed the illustrative limits.
Liquidity and exit	Large caps exit in under 0.1 day at 5% participation; the small-cap takes 0.9 day normally and 17.8 days in stress.
Scenarios and reverse stress	Risk-off −30.9%, stablecoin depeg −2.4%, reverse stress −47.5%; the reverse path breaches the 35% loss floor.
Decision record	Review the small-cap; enforce the 30% single-position cap; classify USDC as counterparty-bearing liquidity; state falsifiers and a review date.
Visual QA	Eight charts; one finding per chart; source/date, unit, and limitation on every figure; bars start at zero; no 3D, pie, or unlabeled dual axis.
Read-only boundary	No wallet, key, signature, approval, bridge, deposit, or order; no transaction was executed.

> **What would change the view:** small-cap exit time below five days, beta below 1.0, or a verified cost basis that changes the return attribution. Until then, the correct output is a conditional review, not a trade.